

P2: Evaluating the efficacy of explanation methods in uncovering spurious correlations

ANIRUDH NAKRA and HENOK GEBEYEHU

1 INTRODUCTION

Modern day machine learning tools often exploit non-interpretable features from the train dataset. In addition to achieving state-of-the-art accuracy on benchmarks, there is also an underlying need to understand which concepts are being learned by the designed model. A natural consequence to designing without such robustness in mind is the fact that a model might learn from unwanted micro-signals that might be unexplainable through a human lens. In the context of Natural Language based tasks, spurious correlations often come from unexpected sources. Adversaries that flip decisions can be generated at a character level without changing the semantic content. [Ebrahimi et al. 2018a],[Ebrahimi et al. 2018b] Spurious correlations can also arise due to biased datasets. Models trained on such corpi are prone to learning unwanted correlations even though the model itself might be robust in theory.[Zhang et al. 2018], [Bolukbasi et al. 2016]. A great practical example of a spurious correlation arisen through biases in datasets is given by [Wang and Culotta 2020] who find that words like *Spielberg* and *New York Subway* have high co-occurences with the sentiment of movie reviews.

We adopt the generic framework suggested in [Wang et al. 2021] for identifying and mitigating spurious correlations.

- (1) **Important Token Identification:** Using the underlying basic causal mapping between the input tokens and the output class in a text classification problem, we evaluate different explainability techniques (such as LIME, SHAP, etc.) and regimes (such as local vs global explanations, post-hoc vs learned explanations) to identify how effective they are in recovering spurious correlations.
- (2) **Identifying spurious correlations:** We adopt the approach outlined by [Wang et al. 2021] and use a cross-dataset consistency approach with the goal being identifying globally meaningful tokens (out of domain importance) and locally meaningful tokens (in domain importance).

This is of particular importance to NLP tasks where we can have a lot of unintended bias built into the dataset due to broad web crawling based collection procedures. We are explaining the models underlying working by uncovering data that might be noise in the sense that researchers and broadly humans think these data samples are not useful for the task its being used for. In this study, we gear the explanations towards Machine Learning researchers.

2 BACKGROUND AND RELATED WORK

Several approaches to tackling such problems have been explored in the literature with solutions such as dataset augmentation [Wu et al. 2022] and adding adversarial examples to training corpus [Jia and Liang 2017] but these efforts are focused on modifying the dataset. These techniques can be great when taken into consideration for generating new datasets, but are infeasible for already existing datasets considering the vast quantity of well-established datasets that are currently available. Other works that explored spurious correlations for NLP focused on training both a



This work is licensed under a Creative Commons Attribution 4.0 International License.

biased model and a debiased model that could confirm if particular input tokens were spurious in nature [He et al. 2019] [Clark et al. 2019], but the limitation of these approaches are that they required spurious tokens to be pre-defined and could not be used to identify new spurious correlations. Currently, the best method for automatically identifying spurious correlations in NLP was proposed by [Wang et al. 2021]. They first identify which tokens in a sentence are considered important for a machine learning model to accomplish a certain task using attention scores. Then they perform a cross-dataset analysis to categorize each important token as either “genuine” (tokens necessary for the model to make an accurate prediction) or “spurious” (tokens that are useful in the training data, but are unreliable in general). The intuition behind the cross-dataset analysis is that if an input token has high attention scores from one dataset and low attention scores in another, then it is likely spurious, but if it has high attention scores in both datasets, then it is likely genuine.

The current limitation of this approach is that the identification of important tokens is solely done using transformer attention scores. It is still unclear whether attention can be considered explanation as summarized by [Bibal et al. 2022], and one of the major points from [Jain and Wallace 2019] was that “there are poor correlations between attention weights and gradient-based or leave-one-out methods for explanation.” This implies that different input saliency techniques will select different input tokens as explanation. So by only utilizing a single input saliency method in the first stage of their framework, it is possible that some spurious correlations remain unidentified.

3 PROPOSAL

For our project, we will adapt the framework developed by [Wang et al. 2021] to identify spurious correlations. We will modify the important token extraction process of their framework by replacing attention scores with various input saliency techniques. The goal is to identify whether using different saliency techniques in the important token extraction stage can uncover new spurious correlations. There is also a concern that the cross-dataset analysis may identify domain-specific tokens as spurious. We will expand on the cross-dataset analysis performed by [Wang et al. 2021] by performing a cross-dataset analysis on datasets from both the same domain as well as different domains to see if context-specific tokens can be properly identified as genuine tokens. We formalize our objectives by posing the following research questions:

RQ1: Are current feature importance and attribution methods effective in identifying important tokens for spurious token identification ?

RQ1b: How does the effectiveness of discovering spurious features vary across different paradigms including gradient vs non gradient based methods, post-hoc vs learning based methods, local vs global methods

RQ2: Can cross dataset consistency reveal different spurious correlations depending on dataset context?

RQ3: Do human users or experts (depending on domain specification) find important tokens actually important?

The novelty in our approach is that we consider more techniques for the process of identifying important tokens. As stated before attention weights have poor correlation with other explanation techniques, and so we believe our technique will be successful because our approach does not make the assumption that attention scores will identify all important tokens to a model’s prediction. By considering more explanation techniques, we expand the possibility of identifying spurious correlations that may have been undetected in the original study.

If we are successful, machine learning researcherse will be the ones who benefit the most because they will be able to identify which features of their dataset are causing their models to be less

generalizable. If spurious correlations are identified and their effects are mitigated, then the model will be more robust whenever it encounters a new dataset that has a distributional shift from the data that was used to train the model. This has the potential significantly improve perceived trust in algorithms for decision-making tasks because models will be making decisions based only on reliable features.

Based on the taxonomy given by [Doshi-Velez and Kim 2017], we adopt the use of a functionally grounded approach. In particular, we will be evaluating on a proxy task which has been shown to be human grounded through the user studies done in [Wang et al. 2021]. As written in the previous sections, we assume for the purpose of the pilot study that the observations of [Wang et al. 2021] have merit and have some epsilon error/misalignment with respect to the human grounded evaluation done in the paper. So, further evaluation would then consider an offset of epsilon as assumed in the study and the evaluation will be done completely w.r.t the references in [Wang et al. 2021]. Our evaluation outline is as follows:

- (1) **Type of evaluation:** Functionally grounded
- (2) **Type of task:** Proxy task
- (3) **Type of application:** Prototype
- (4) **Type of deployment:** Lab based (assumed due to [Wang et al. 2021])

Our project structure is as follows:

- (1) **Task:** Text classification (Sentiment analysis)
- (2) **Data:** For answering RQ1: SST-2, Yelp reviews. For answering RQ2 and RQ3, Movie Rationales as well
- (3) **Metrics:** Based on the research questions, the evaluation approach wants to calculate the alignment between important tokens found using the different techniques and the reference tokens. We aim to evaluate different techniques with a score that computes how well we are able to recover the same important tokens that were found in [Wang et al. 2021]. Using this idea, a possible metric would be as follows

- **Intersection over Union: (IoU)**

$$\text{IoU} = \frac{1(A \text{ AND } B)}{1(A \text{ OR } B)}$$

where **1** represents the count function, A represents the technique being evaluated and B represents our reference tokens. Further, it might not be good to punish a technique if it is able to recover more genuinely spurious tokens. To do so, we propose a traditional accuracy based idea:

- **Recovery percentage: (RP)**

$$\text{RP} = \frac{1(A \text{ AND } B)}{1(B)} * 100$$

where **1** represents the count function, A represents the technique being evaluated and B represents our reference tokens.

For RQ2, we want to calculate the alignment between the found spurious tokens and the reference spurious tokens. For RQ3, we want to calculate the alignment between the human annotations (what human considers is actually relevant to the task) and the tokens we find are genuine and spurious. Both of these can be analysed using IoU and RP with the desired change in B to be tokens relevant to the respective RQ.

In our pilot study, the conditions we compare for each research question are as follows:

- (1) **RQ1:** Comparison of important token identification across 5 different highlight/saliency techniques i.e Attention (Learned technique, Local technique), Logistic Regression (Learned technique, Global technique), LIME (Post-hoc technique, Local technique), Gradient Norm (Gradient based technique, Local technique) and Gradient x Input (Gradient based technique, Local technique)

- (2) **RQ2:** Comparison of spurious tokens with references in [Wang et al. 2021] would involve solving for the alignment using IoU or RP of some methods from RQ1 with [Wang et al. 2021]. Further this would involve computing alignment scores when doing cross-dataset analysis within and out of domain.
- (3) **RQ3:** Comparison of human annotations with spurious tokens would involve solving for the alignment using IoU or RP of some methods from RQ1 with annotations given in the Movie Rationales dataset. The lower IoU and RP, the better.

The tables to be filled as the result of our experiment are: (The table for RQ1b will involve averaging scores of RQ1 results from the 5 techniques to form averaged scores for the regimes)

Problem Setup	Metric	Attention	LIME	Log. Reg.	GradNorm	GradxInp
RQ1	IoU	-	-	-	-	-
	RP	-	-	-	-	-
RQ2	IoU	-	-	-	-	-
	RP	-	-	-	-	-
RQ3	IoU	-	-	-	-	-
	RP	-	-	-	-	-

Table 1. For RQ1-3

Metric	Global	Local	Gradient based	Learned	Post-hoc
IoU	-	-	-	-	-
RP	-	-	-	-	-

Table 2. For RQ1b

4 LOGISTICS

4.1 Cost

Our experiments require no crowdsourcing. Fine-tuning a model and extracting saliency scores with the various techniques will require access to a GPU. For this purpose, we will use Google Colab, but if it doesn’t provide enough compute, we would use servers with GPU VRAM > 2Gb.

4.2 Timeline

Steps for completion:

- (1) Extract important tokens using attention scores as done in [Wang et al. 2021]
- (2) Generate a list of candidate spurious tokens by filtering out the genuine tokens from the list of important tokens obtained in the previous step. This will be done by performing a Cross-Dataset Analysis using the following datasets:
 - SST-2 (Stanford Tree Bank - 2)
 - Yelp reviews
 - Movie Rationales
- (3) Repeat steps 1 and 2 for the following saliency techniques:
 - Logistic regression
 - LIME
 - Gradient norm
 - Gradient x Input
- (4) Analyze the spurious tokens obtained from using each saliency technique to answer each research question

We have already completed step 1 and parts of step 2. We can reasonably expect to complete the first three steps by April 21st as our midterm deliverable. After steps 1-3 are complete, step 4 can be completed by May 8th as our final deliverable.

REFERENCES

- Adrien Bibal, Rémi Cardon, David Alfter, Rodrigo Wilkens, Xiaou Wang, Thomas François, and Patrick Watrin. 2022. Is Attention Explanation? An Introduction to the Debate. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Dublin, Ireland, 3889–3900. <https://doi.org/10.18653/v1/2022.acl-long.269>
- Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. 2016. Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. <https://doi.org/10.48550/ARXIV.1607.06520>
- Christopher Clark, Mark Yatskar, and Luke Zettlemoyer. 2019. Don’t Take the Easy Way Out: Ensemble Based Methods for Avoiding Known Dataset Biases. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics, Hong Kong, China, 4069–4082. <https://doi.org/10.18653/v1/D19-1418>
- Finale Doshi-Velez and Been Kim. 2017. Towards A Rigorous Science of Interpretable Machine Learning. arXiv:1702.08608 [stat.ML]
- J. Ebrahimi, Daniel Lowd, and Dejing Dou. 2018a. On Adversarial Examples for Character-Level Neural Machine Translation. In *International Conference on Computational Linguistics*.
- Javid Ebrahimi, Anyi Rao, Daniel Lowd, and Dejing Dou. 2018b. HotFlip: White-Box Adversarial Examples for Text Classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Association for Computational Linguistics, Melbourne, Australia, 31–36. <https://doi.org/10.18653/v1/P18-2006>
- He He, Sheng Zha, and Haohan Wang. 2019. Unlearn Dataset Bias in Natural Language Inference by Fitting the Residual. arXiv:1908.10763 [cs.CL]
- Sarthak Jain and Byron C. Wallace. 2019. Attention is not Explanation. arXiv:1902.10186 [cs.CL]
- Robin Jia and Percy Liang. 2017. Adversarial Examples for Evaluating Reading Comprehension Systems. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Copenhagen, Denmark, 2021–2031. <https://doi.org/10.18653/v1/D17-1215>
- Tianlu Wang, Rohit Sridhar, Diyi Yang, and Xuezhi Wang. 2021. Identifying and Mitigating Spurious Correlations for Improving Robustness in NLP Models. <https://doi.org/10.48550/ARXIV.2110.07736>
- Zhao Wang and Aron Culotta. 2020. Identifying Spurious Correlations for Robust Text Classification. (2020). <https://doi.org/10.48550/ARXIV.2010.02458>
- Yuxiang Wu, Matt Gardner, Pontus Stenetorp, and Pradeep Dasigi. 2022. Generating Data to Mitigate Spurious Correlations in Natural Language Inference Datasets. <https://doi.org/10.48550/ARXIV.2203.12942>
- Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell. 2018. Mitigating Unwanted Biases with Adversarial Learning. <https://doi.org/10.48550/ARXIV.1801.07593>